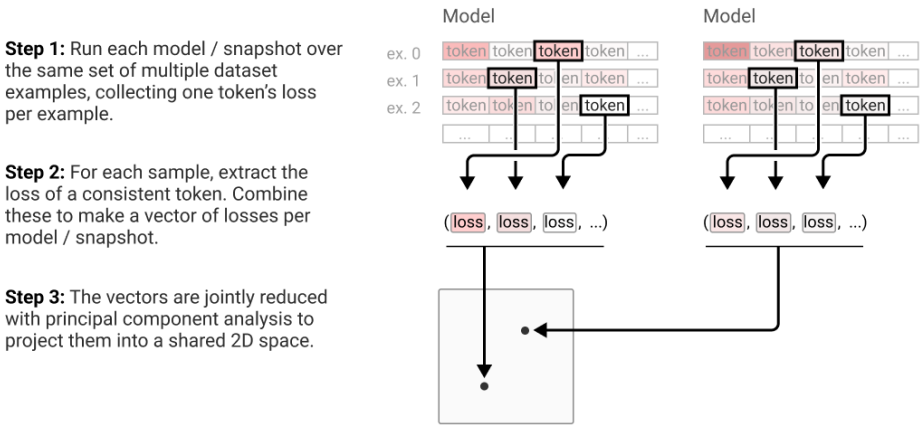
We start with a collection of models. (In our use, we'll train several different model architectures, saving dozens of “snapshots” of each over the course of training. We’ll use this set of snapshots as our collection of models.) Next, we collect the log-likelihoods each model assigns to a consistent set of 10,000 random tokens, each taken from a different example sequence. We combine these log-likelihoods into a "per-token loss vector" and apply Principal Component Analysis (PCA):



A more detailed discussion of technical details can be found in the Appendix.

By applying this method to snapshots over training for multiple models, we can visualize and compare how different models' training trajectories evolve in terms of their outputs. Since we're using PCA, each direction can be thought of as a vector of log-likelihoods that models are moving along. We particularly focus on the first two principal components, since we can easily visualize those. Of course, models also move in directions not captured by the first two principal components, but it's a useful visualization for capturing the highest-level story of training.

Arguments that induction heads are the mechanism for the majority of in-context learning.

Now we’ll proceed to the main part of the paper, which makes the case that induction heads may *provide the primary mechanism* for the majority of in-context learning for transformer models in general. As stated in the introduction, this is a very broad hypothesis and much of our evidence is indirect, but nevertheless we believe that all the lines of evidence together make a relatively strong, though not conclusive, case.

Before we go through the arguments, it’s useful to delineate where the evidence is more conclusive vs. less conclusive. This is shown in the table below. For small, attention-only models, we believe we have strong evidence that attention heads are the mechanism for the majority of in-context learning, as we have evidence supported by ablations and mechanistic reverse engineering. Conversely, for all models, we can make a strong case that induction heads play *some role* in in-context learning, as we can demonstrate examples and show suggestive correlations. However, the larger the models get, the harder it is to establish that induction heads account for the actual *majority* of in-context learning. For large models with MLP’s, we must therefore rely on mainly correlational evidence, which could be confounded. We explore alternate hypotheses throughout, including at the end of Argument 1 and again briefly in Argument 6.

SUMMARY OF EVIDENCE FOR SUB-CLAIMS (STRONGEST ARGUMENT FOR EACH)

	Small Attention-Only	Small with MLPs	Large Models
Contributes Some	Strong, Causal	Strong, Causal	Medium, Correlational & Mechanistic
Contributes Majority	Strong, Causal	Medium, Causal	Medium, Correlational

Here is the list of arguments we'll be making, one per section, repeated from the introduction:

- **Argument 1** *(Macroscopic co-occurrence)***:** Transformer language models undergo a "phase change" early in training, during which induction heads form and simultaneously in-context learning improves dramatically.
- **Argument 2** *(Macroscopic co-perturbation):* When we change the transformer architecture in a way that shifts whether induction heads can form (and when), the dramatic improvement in in-context learning shifts in a precisely matching way.
- **Argument 3** *(Direct ablation):* When we directly "knock out" induction heads at test-time in small models, the amount of in-context learning greatly decreases.
- **Argument 4** (*Specific examples of induction head generality*): Although we define induction heads very narrowly in terms of copying literal sequences, we empirically observe that these same heads also appear to implement more sophisticated types of in-context learning, including highly abstract behaviors, making it plausible they explain a large fraction of in-context learning.
- **Argument 5** (*Mechanistic plausibility of induction head generality):* For small models, we can explain mechanistically how induction heads work, and can show they contribute to in-context learning. Furthermore, the actual mechanism of operation suggests natural ways in which it could be repurposed to perform more general in-context learning.
- **Argument 6** (*Continuity from small to large models*): In the previous 5 arguments, the case for induction heads explaining in-context learning is stronger for small models than for large ones. However, many behaviors and data related to both induction heads and in-context learning are smoothly continuous from small to large models, suggesting the simplest explanation is that mechanisms are the same.

For each argument, we'll have a similar table to the one in this section, showing the strength of the evidence provided by that claim as it applies to large/small models and some/most of context learning. The table above is the sum of the evidence from all six lines of reasoning.

Argument 1: Transformer language models undergo a "phase change" during training, during which induction heads form and simultaneously in-context learning improves dramatically.

Our first line of evidence comes from correlating measures of in-context learning and measures of induction head presence over the course of training. Specifically, we observe a tight co-variation between them across dozens of models, of different sizes, trained on different datasets (See Model Analysis Table for more on the models in which we observe this co-occurrence.).

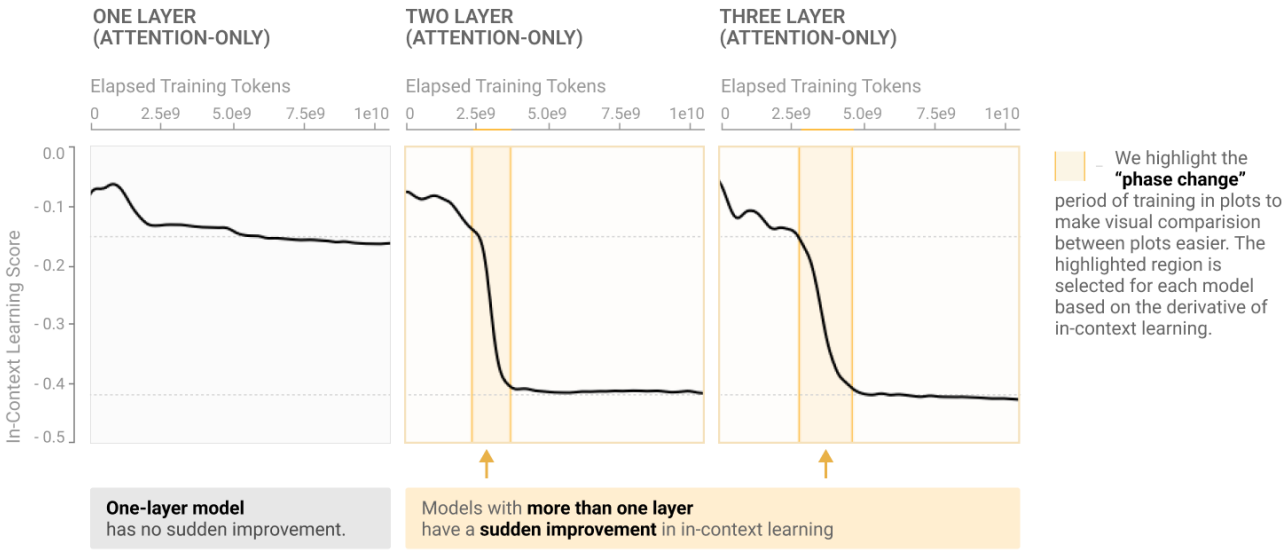
The table below summarizes the quality of this evidence for the models we have studied: it applies to both large and small models, and is the expected outcome if induction heads were responsible for the majority of in-context learning, but it is only correlational and so could be confounded (discussed more below).

STRENGTH OF ARGUMENT FOR SUB-CLAIMS

	Small Attention-Only	Small with MLPs	Large Models
Contributes Some	Medium, Correlational	Medium, Correlational	Medium, Correlational
Contributes Majority	Medium, Correlational	Medium, Correlational	Medium, Correlational

Our first observation is that if we measure in-context learning for transformer models over the course of training (defined as the 50th token loss minus the 500th token loss as described in Key Concepts), it develops abruptly in a narrow window early in training (roughly 2.5 to 5 billion tokens) and then is constant for the rest of training (see figure below). Before this window there is less than 0.15 nats of in-context learning, after it there is roughly 0.4 nats, an amount that remains constant for the rest of training and is also constant across many different model sizes (except for the one layer model where not much in-context learning ever forms). This seems surprising – naively, one might expect in-context learning to improve gradually over training, and improve with larger model sizes,[9] as most things in machine learning do.

MODELS WITH MORE THAN ONE LAYER HAVE AN ABRUPT IMPROVEMENT IN IN-CONTEXT LEARNING



Although we only show three models above, the pattern holds true very generally: many examples are shown in the Model Analysis Table later in the paper, including models of varying model architecture and size.